

Wilson, Chow Sin Long

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Full-Stack Software Engineer · AWS Certified Solutions Architect – Associate

SUMMARY

Full-stack **software engineer** with **5 years of experience** delivering **15+** React and React Native products from Figma through Node.js APIs and AWS, including Apple App Store and Google Play releases. Cut load times by **80%** and search latency by **90%**. Now leading a multi-channel **Generative AI** platform with durable workflows, memory, RAG, and evals. **AWS Certified Solutions Architect – Associate**.

EXPERIENCE

Software Developer, Vioo Company Limited

May 2022 – Present

Primary engineer for concurrent client products, owning delivery from UX scoping through cloud deployment.

- Shipped and maintained **15+** React / React Native apps with high-fidelity Figma design, shared design systems, and Apple App Store / Google Play releases.
- Migrated WordPress sites to Strapi / Sanity and Meilisearch / Algolia in a **Turborepo** on **SST** without interrupting publishing; cut load times by **80%**, search latency by **90%**, and achieved PageSpeed scores above **95**.
- Built typed Node.js APIs with Hono, OpenAPI, and tRPC on Lambda / API Gateway and **ECS Fargate**; modeled org-scoped Postgres with Drizzle / Prisma, Better Auth, OAuth 2.0 / OIDC, and RBAC.
- Owned shared **Radix** / **Tailwind** / **shadcn/ui** component and hook APIs; **Storybook**, Fumadocs, and MDX guidance eliminated duplicate UI across client apps.
- Set **Vitest**, React Testing Library, and **Playwright** / **Cypress** quality gates for shared UI and critical flows.
- Instrumented web, mobile, and API releases with **Sentry** for production crashes, device issues, and API errors.
- Adopted **tRPC** and Scalar-generated API docs, cutting integration time by **50%** and eliminating schema-mismatch errors.
- Built GitHub Actions + **SST** CI/CD with linting, type-checking, tests, and builds; cut deployment time by **50%+**.
- Mentored interns and junior developers through pairing, code reviews, and production delivery.

SELECTED PROJECTS

Multi-Channel Agentic Platform

Apr 2026 – Present

Durable agent platform across web and Discord.

- Made agent runs durable streams so clients could disconnect, reconnect, and resume without losing progress.
- Built Mastra / AI SDK workflows, memory, schedules, tools, pause/resume, retries, sandboxing, and skills.
- Shipped org-scoped RAG (ingestion, chunking, Pinecone query/reranking) and eval-gated releases.
- Ran Hono + Better Auth on **ECS Fargate**, with Postgres, S3, and non-blocking ClickHouse traces for cost/latency.
- Shipped chat, tool-invocation, multi-step flows, and TipTap @mentions for attaching workspace files.

Healthcare AI Agent

July 2025 – Apr 2026

Healthcare and shopping assistant grounded in uploaded documents and product data.

- Owned document upload, questionnaires, product/cart actions, and admin flows; Better Auth, HttpOnly cookies, and RBAC separated client/admin roles.
- Split admin list state: URL-owned filters/pagination, **TanStack Query** server cache, and **TanStack Table** presentation.
- Built document ingestion and scoped RAG so recommendations stayed grounded in the uploaded report.

High-Performance Editorial Platform

Nov 2025

Strapi, Meilisearch, and **Next.js** editorial platform in the WordPress-to-headless migration.

- Reserved image space and optimized caching/images: **LCP < 0.5s**, **CLS < 0.1**, PageSpeed **>90** (lab).
- Used hybrid ISR: main pages/latest 100 articles at build, rest on demand; Strapi **GraphQL**, **Next.js** cache tags, and webhooks revalidated publishes without full rebuilds.
- Made keyboard navigation, accessible form errors, semantics, and Lighthouse/axe checks release requirements.
- Split interactive state from page data: **XState** + **TanStack Form** / **Zod**, route-owned locale, and hybrid-ISR content.
- Built a Strapi plugin for AI-assisted content generation with AI SDK.

React Native Membership Platform

shipped Apr 2024

- Owned digital card, points, and offers from Figma through Apple App Store / Google Play release; maintained the Express API and **Next.js** admin.
- Stored refresh tokens in **SecureStore** / Keychain, gated sessions with biometrics, and generated typed **OpenAPI** REST clients.
- Maintained **99%** crash-free sessions by diagnosing crashes, device issues, and API bottlenecks with **Sentry**.
- Fixed JS-thread feed stalls with **FlashList**, memoized rows, smaller cached images, and native-thread **Reanimated** / **Moti**; froze/deferred inactive tabs, trading later first paint for **60 FPS** (RN Perf Monitor, lab).
- Routed FCM promo/milestone deep links to the matching screen.

OTHER WORK

Also shipped React Native student and restaurant loyalty apps, a Flutter elderly-care app, Shopify storefronts, CMS sites, a Skia photobooth, and multiple RBAC admin systems.

SKILLS

Languages: TypeScript, Node.js, SQL

Frontend: React, Next.js, React Native, Expo, HTML/CSS, Tailwind CSS, shadcn/ui, Figma, TipTap

UI behavior: TanStack Query, TanStack Form, React Hook Form, XState, TanStack Table, next-intl

Architecture: state modeling (URL/server/local), component APIs, routing, SSR/ISR, GraphQL, tRPC, OpenAPI clients, Storybook, Fumadocs

Backend: Hono, Drizzle, Prisma, Postgres, DynamoDB, Better Auth

Auth & security: OAuth 2.0/OIDC, JWT, HttpOnly cookies, CORS, RBAC

AI: Mastra, Vercel AI SDK, LangChain, Pinecone, Strands, AWS Bedrock, evals, ClickHouse tracing

Quality: Vitest, Playwright, Cypress, React Testing Library, Lighthouse, axe, Sentry

Mobile: FlashList, Reanimated, Moti, Skia, Unistyles, SecureStore/biometrics, FCM

AWS & DevOps: SST, Docker, ECS Fargate, Lambda/API Gateway, CloudFront/CDN, GitHub Actions, AWS, Turborepo

QUALIFICATIONS

- AWS Certified Solutions Architect – Associate, 2026
- AWS Certified Cloud Practitioner, 2025

EDUCATION

University College London

2016 – 2020

- Master of Pharmacy (MPharm) — First Class Honours
- Transitioned from pharmacy to software development after finding passion in building applications.